

Data and Decisions

Fall 2026

One Population, Many Samples

In-Class Team Assignment

SOLUTION

From the bCourses Class 2 page download “Airline Ticket Prices.xlsx”, “fares_sample_1.csv”, and “fares_sample_2.csv”. Use any AI tool that you wish. The big file (Airline Ticket Prices.xlsx) is 10,266 recent roundtrip JFK–LAX tickets — and in this exercise it is the entire POPULATION: $\mu = \$368.79$, $\sigma = \$179.72$. Here you have the population of interest (i.e. truth), so you can check everything your AI analyst tells you.

1. Spot the biased sample

The two CSVs are samples of 1,000 tickets each that a colleague pulled from the population. One is a simple random sample; one is not. Paste both into your AI: “Compare these two samples. Which one looks like a random sample of all JFK–LAX tickets, and which doesn’t? What selection rule produced the other one?”

- Which file is the random sample? What was the selection rule behind the other one, and which column gives it away?

Solution: fares_sample_2.csv is the SRS (mean fare \$360.23, close to $\mu = \$368.79$; composition matches the population). fares_sample_1.csv contains only tickets bought 7 or fewer days in advance — the advance column maxes out at 7, and the composition shifts with it (Saturday-stayover share 9% vs. 23%, business class 5% vs. 2%). Its mean fare is \$511.57.

- If you had used the biased sample to estimate the average fare, how far off would you be, and in which direction? What real-world survey mistake is this the same as?

Solution: About \$143 too high (\$511.57 vs. \$368.79) — late bookers pay walk-up fares. Same structure as any frame/selection error: surveying only the customers easiest to reach (people at the airport counter, respondents to a pop-up, landline answerers) and calling it “the average customer.” More data from this frame would not fix it.

2. How much variability is there in the “sample mean” for tickets?

Now use the full population file. Ask your AI assistant: “Draw 500 random samples of $n = 100$ tickets, compute each sample’s mean fare, and report the standard deviation of the 500 means. Plot them.”

- Using only μ , σ , and n , what should the standard deviation of those 500 sample means be close to? Does the simulation check out?

Solution: $\sigma/\sqrt{n} = 179.72/\sqrt{100} \approx \17.97 ; the simulated standard deviation should land within a dollar or so of that. Remember, the sample mean has a sampling distribution, and AI can show / simulate it.

- Would your answer change if the airline had sold 10 million tickets instead of 10,266? One sentence answer.

Solution: No — the standard error depends on σ and n , not on the population size. (The same reason J.D. Power doesn't need a bigger sample in a year when more cars are sold.)

3. Set your control limits

You are the route's revenue manager. Fares are set by an automated pricing system, and pricing systems break quietly — a fare class filed wrong, a competitor-matching bug, cheap seats left on sale too long. (Airlines have accidentally sold transatlantic tickets for \$75 this way.) You can't inspect every ticket, so each week you audit a fresh random sample of $n = 100$ of that week's tickets. When the system is working as designed, fares are drawn from the population above — so the weekly sample mean behaves exactly like the distribution you simulated in part 2: centered at \$368.79 with $SE \approx \$17.97$. If revenue drifts too far from center, in either direction, something may be broken. This week's sample averaged \$412.

- e. Set control limits for the weekly audit at an α you choose. Justify the choice in one sentence each on the cost of a false alarm (Type I) and the cost of a missed problem (Type II).

Solution: $SE = \$17.97$. At $\alpha = .05$: $368.79 \pm 1.96(17.97) = [\$333.60, \$404.00]$. At $\alpha = .01$: $368.79 \pm 2.58(17.97) = [\$322.40, \$415.20]$. Either is defensible (a false alarm could waste an analyst-day of investigation; a missed pricing bug could leak revenue for weeks).

- f. Investigate this week or not? Note: your answer may legitimately differ from the next team's — explain why in one sentence.

Solution: \$412 is $z = (412 - 368.79)/17.97 \approx 2.40$ standard errors on either side: OUTSIDE the 95% limits but INSIDE the 99% limits. Teams that priced false alarms as cheap ($\alpha=0.05$) investigate; teams that priced them as expensive ($\alpha=0.01$) wait a week. Note, if you run this audit on 20 routes every week at $\alpha = .05$, the chance of at least one false alarm per week is $1 - .95^{20} \approx 64\%$ (because of repeated testing accumulating type I errors).

Before you hand this in: add your team's AI-use statement (a few sentences — which tools you used and what you used them for).